

MTU_ValueService **Technical Documentation**

Diesel engine
12 V 4000 P61, P81
16 V 4000 P61, P81
Application group 3B

Operating Instructions
MS15003/02E



Printed in Germany

© 2011 Copyright MTU Friedrichshafen GmbH

This Publication is protected by copyright and may not be used in any way whether in whole or in part without the prior written permission of MTU Friedrichshafen GmbH. This restriction also applies to copyright, distribution, translation, microfilming and storage or processing on electronic systems including data bases and online services.

This handbook is provided for use by maintenance and operating personnel in order to avoid malfunctions or damage during operation.

Subject to alterations and amendments.

Commissioning Note

Important

Please complete and return the “Commissioning Note” card below to MTU Friedrichshafen GmbH.

The Commissioning Note information serves as a basis for the contractually agreed logistic support (warranty, spare parts, etc.).



Postcard
MTU Friedrichshafen GmbH
Technical Information Management
Dept. VOT
88040 Friedrichshafen
GERMANY

Commissioning Note

Please use block capitals!



Engine No.:	MTU works order no.:
Engine model:	Date put into operation:
Application : * ☐ Marine ☐ Rail ☐ Genset ☐	Type : Manufacturer:
End user`s address:	
Remarks:	

**Commissioning
Note**

Tick appropriate box

1	Safety	9
1.1	General conditions	9
1.2	Personnel and organizational requirements	10
1.3	Transport.....	11
1.4	Crankshaft transport locking device	12
1.5	Safety regulations for startup and operation.....	15
1.6	Safety regulations for maintenance and repair work.....	16
1.7	Auxiliary materials, fire prevention and environmental protection	18
1.8	Conventions for safety instructions in the text.....	20
1.9	ATEX (where applicable).....	21
2	Product Summary.....	23
2.1	Plant Overview.....	23
2.1.1	Use.....	23
2.1.2	ECS-5 – Use of devices.....	31
2.1.3	MDEC – Functions	34
2.1.4	Peripheral Interface Modules (PIMs).....	39
2.1.5	Printed circuit board SCB 3.....	41
2.1.6	LOP – Controls	44
2.2	Engine Layout.....	46
2.2.1	Engine layout.....	46
2.2.2	Sensors, actuators and injectors – Overview.....	47
2.3	Engine Side and Cylinder Designations.....	52
2.3.1	Engine side and cylinder designations.....	52
2.4	Main Engine Dimensions.....	53
2.4.1	Main engine dimensions.....	53
2.5	Firing Sequence.....	54
2.5.1	Firing order.....	54
2.6	Technical Data.....	55
2.6.1	12/16V 4000 Px1 engine data.....	55
2.6.2	SAM.....	58
2.6.3	SCB 3 – Technical data	62
3	Operation.....	63
3.1	Putting the engine into operation after extended out-of-service periods (>3 months).....	63
3.2	Putting the engine into operation after scheduled out-of-service-period	64
3.3	Start engine in manual mode (testing mode).....	65
3.4	Starting the engine in emergency situations (override mode).....	66
3.5	Operational checks.....	67
3.6	Stop engine in manual mode (testing mode).....	68

3.7	Emergency stop.....	69
3.8	After stopping the engine – engine remains ready for operation.....	70
3.9	After stopping the engine – putting the engine out of service.....	71
4	Maintenance.....	73
4.1	Maintenance task reference table [QL1].....	73
5	Troubleshooting.....	75
5.1	Troubleshooting.....	75
5.2	Fault indication on printed circuit board SAM.....	78
6	Task Description.....	89
6.1	Engine.....	89
6.1.1	Engine – Barring manually.....	89
6.1.2	Engine – Barring with starting system.....	90
6.1.3	Engine test run.....	91
6.2	Cylinder Liner.....	92
6.2.1	Cylinder liner – Endoscopic examination.....	92
6.2.2	Instructions and comments on endoscopic and visual examination of cylinder liners.....	94
6.3	Crankcase Breather.....	95
6.3.1	Crankcase breather – Oil separator replacement, diaphragm check and replacement.....	95
6.4	Valve Drive.....	97
6.4.1	Valve gear – Lubrication.....	97
6.4.2	Valve clearance – Check and adjustment.....	98
6.4.3	Cylinder head cover – Removal and installation.....	101
6.5	Injection Pump / HP Pump.....	102
6.5.1	HP fuel pump – Relief bore check.....	102
6.6	Injection Valve / Injector.....	103
6.6.1	Injector – Replacement.....	103
6.6.2	Injector – Removal and installation.....	104
6.7	Fuel Filter.....	110
6.7.1	Fuel filter – Replacement.....	110
6.7.2	Fuel prefilter – Differential pressure check and adjustment of gauge.....	112
6.7.3	Fuel prefilter – Draining.....	113
6.7.4	Fuel prefilter – Flushing.....	115
6.7.5	Fuel prefilter – filter element replacement.....	117
6.8	Charge-Air Cooling General, Left-Hand Side.....	119
6.8.1	Intercooler – Checking condensate drains for coolant discharge and obstructions.....	119
6.9	Air Filter.....	120
6.9.1	Air filter replacement.....	120
6.9.2	Air filter – Removal and installation.....	121
6.10	Air Intake.....	122

6.10.1	Service indicator – Signal ring position check	122
6.10.2	Emergency air shutoff flaps – Lubrication	123
6.10.3	Emergency air-shutoff flaps – Functional check with electric actuation	124
6.11	Starting Equipment	125
6.11.1	Air starter – Manual operation	125
6.12	Lube Oil System, Lube Oil Circuit	126
6.12.1	Engine oil – Change	126
6.12.2	Engine oil – Level check	128
6.12.3	Engine oil – Sample extraction and analysis	129
6.13	Oil Filtration / Cooling	130
6.13.1	Engine oil filter – Replacement	130
6.13.2	Centrifugal oil filter – Cleaning and filter-sleeve replacement	133
6.14	Coolant Circuit, General, High-Temperature Circuit	135
6.14.1	Engine coolant – Filling	135
6.14.2	Engine coolant – Draining	137
6.14.3	Engine coolant – Change	138
6.14.4	Engine coolant – Level check	139
6.14.5	Engine coolant pump – Relief bore check	140
6.14.6	Engine coolant – Sample extraction and analysis	141
6.15	Low-Temperature Circuit	142
6.15.1	Charge-air coolant – Filling	142
6.15.2	Charge-air coolant – Draining	145
6.15.3	Charge-air coolant – Change	147
6.15.4	Charge-air coolant – Level check	148
6.15.5	Charge-air coolant pump – Relief bore check	149
6.16	Wiring (General) for Engine/Gearbox/Unit	150
6.16.1	Engine wiring – Check	150
6.17	Accessories for (Electronic) Engine Governor / Control System	151
6.17.1	Engine governor and connectors – Cleaning	151
6.17.2	Engine governor – Checking plug-in connections	152
6.17.3	EMU and connectors – Cleaning	153
6.17.4	Engine monitoring system – Checking plug connections	155
7	Engine Governor Assignment and Circuitry	157
7.1	Engine Governor	157
7.1.1	Engine governor channel circuitry	157
7.1.2	MDEC governor assignment	167
7.2	Engine Monitoring	169
7.2.1	LOP and connectors – Cleaning	169
7.2.2	LOP – visual inspection	170
7.2.3	LOP – Test procedures	173
7.3	Connector Assignment	175
7.3.1	Connector pin assignment	175
8	Appendix A	185
8.1	Abbreviations	185

	8.2	MTU Contact/Service partners.....	188
9		Appendix B.....	189
	9.1	Special Tools	189
	9.2	Consumables.....	190
	9.3	Index.....	191

1 Safety

1.1 General conditions

General

In addition to the instructions in this publication, the applicable country-specific legislation and other compulsory regulations regarding accident prevention must be observed. This state-of-the-art engine has been designed to meet all applicable laws and regulations. The engine may nevertheless present a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Modifications or conversions
- Non-compliance with the Safety Instructions

Correct use

The engine is intended solely for use in accordance with contractual agreements and the purpose envisaged for it on delivery. Any other use is considered improper use. The engine manufacturer accepts no liability whatsoever for resultant damage or injury in such case. The responsibility is borne by the user alone.

Correct use also includes observation of and compliance with the maintenance specifications.

Modifications or conversions

Unauthorized modifications to the engine represent a safety risk.

MTU will accept no liability or warranty claims for any damage caused by unauthorized modifications or conversions.

Spare parts

Only genuine MTU spare parts must be used to replace components or assemblies. The engine manufacturer accepts no liability whatsoever for damage or injury resulting from the use of other spare parts and the warranty shall be voided in such case.

Reworking components

Repair or engine overhaul must be carried out in workshops authorized by MTU.

1.2 Personnel and organizational requirements

Personnel requirements

All work on the engine shall be carried out by trained and qualified personnel only.

The specified legal minimum age must be observed.

The operator must specify the responsibilities of the operating, maintenance and repair personnel.

Organizational measures

This publication must be issued to all personnel involved in operation, maintenance, repair or transportation.

Keep it handy in the vicinity of the engine such that it is accessible to operating, maintenance, repair and transport personnel at all times.

Use the manual as a basis for instructing personnel on engine operation and repair with an emphasis on explaining safety-relevant instructions.

This is particularly important in the case of personnel which only occasionally performs work on or around the engine.

This personnel must be instructed repeatedly.

For the identification and layout of the spare parts during maintenance or repair work, take photos or use the spare parts catalog.

Working clothes and protective equipment

Wear proper protective clothing for all work.

Use the necessary protective equipment for the given work to be done.